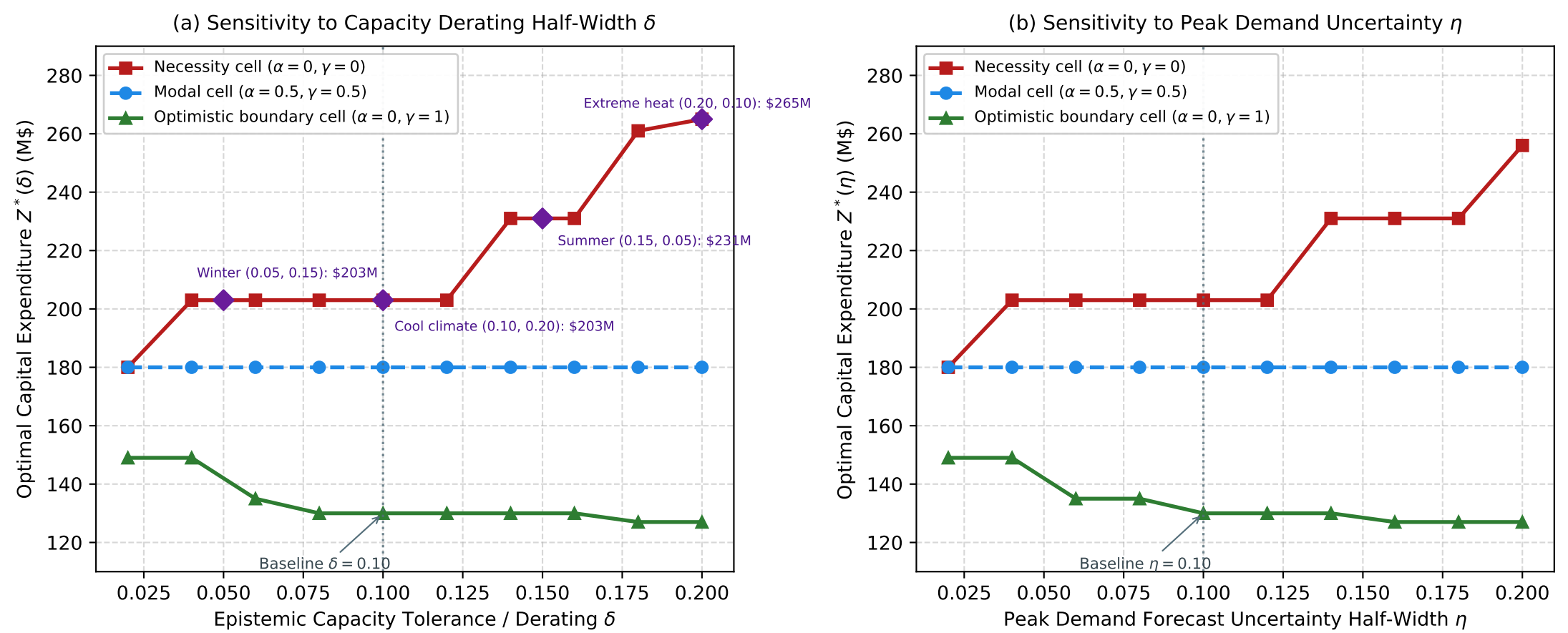
(a) Sensitivity to Capacity Derating Half-Width δ
Necessity cell ($\alpha = 0, \gamma = 0$)
Modal cell ($\alpha = 0.5, \gamma = 0.5$)
Optimistic boundary cell ($\alpha = 0, \gamma = 1$)
Extreme heat (0.20, 0.10): $265M
Summer (0.15, 0.05): $231M
Winter (0.05, 0.15): $203M
Cool climate (0.10, 0.20): $203M
Baseline $\delta = 0.10$
Optimal Capital Expenditure $Z^*(\delta)$ (M$)
Epistemic Capacity Tolerance / Derating δ
(b) Sensitivity to Peak Demand Uncertainty η
Necessity cell ($\alpha = 0, \gamma = 0$)
Modal cell ($\alpha = 0.5, \gamma = 0.5$)
Optimistic boundary cell ($\alpha = 0, \gamma = 1$)
Baseline $\eta = 0.10$
Optimal Capital Expenditure $Z^*(\eta)$ (M$)
Peak Demand Forecast Uncertainty Half-Width η